



Introduction

In the context of database management and analysis, a **union** refers to the process of combining two tables into a single table. Determining whether two tables are **unionable** can depend on multiple factors, including their structure, semantics, and context.

In this study, you will evaluate a series of table pairs and decide whether they are unionable. For each question, you will first provide your own judgment. After that, you will be shown an AI-generated suggestion, which can include a recommendation, an explanation, or a confidence value. The AI support is provided for research purposes and **may not always be correct**.

Please answer each question based on your own reasoning and judgment. Some cases may be straightforward, while others may be more ambiguous. We are interested in **how** people make these decisions and how AI assistance **may affect** that process. Your responses will be collected anonymously and used only for research purposes. Participation is voluntary, and you

may stop the survey at any time. By continuing, you indicate that you understand the purpose of the study and agree to participate.

Before we begin, please provide the following information about you for the purpose of this research study. Please note that all your responses will remain anonymous, and your privacy is fully assured.

Please select your age group:

- ☐ Under 18
- ☐ 18 - 24
- ☐ 25 - 34
- ☐ 35 - 44
- ☐ 45 - 54
- ☐ 55 & above

What is the highest level of education you have completed?

- ☐ High school diploma or equivalent
- ☐ Bachelor's degree
- ☐ Master's degree

- ☐ Doctoral degree
- ☐ No degree
- ☐ Other (please specify):

How would you describe your proficiency in English?

- ☐ Basic
- ☐ Intermediate
- ☐ Proficient
- ☐ Fluent
- ☐ Native speaker

What is your major or primary field of study?

Read before proceeding:

As you move to the main section of the Survey, after each question you will be asked to explain your decision in your own words here, even though it is optional we would like to know why you made that particular decision.

Good Luck!

Main Survey_Pilot_HAI

Do you think Table A and Table B are unionable?

Table A		
Environment	Air	Water
Ice	Biomass	Natural Lighting
Animals	Melting Glaciers	Reduce Trash
Oceans	Heat Waves	Aquaculture
Plants	Wave-Particle Duality	Irrigation
Energy	Solar Tubes	Dikes

Table B		
Environment	Air	Water
Land	Zero Net Energy	Floodplain
Energy	Tubular Skylights	Drainage Basins
Animals	Light Tubes	Channelization
Oceans	Passive Cooling	Reefs
Plants	Daylighting System	Flood Planning

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your answer to the question?

0 10 20 30 40 50 60 70 80 90 100

Confidence Level

Do you think Table A and Table B are unionable?

Table A		
Environment	Air	Water
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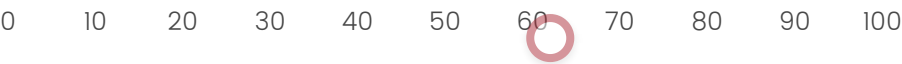
Table B		
Environment	Air	Water
Land	Zero Net Energy	Floodplain
Energy	Tubular Skylights	Drainage Basins
Animals	Light Tubes	Channelization
Oceans	Passive Cooling	Reefs
Plants	Daylighting System	Flood Planning

LLM Recommendation: Unionable

After reviewing the AI suggestion, what is your final answer?

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your final answer?



Confidence Level

Please briefly explain your final decision and describe whether the AI suggestion changed, confirmed, or did not affect your reasoning.

How helpful was the AI suggestion for this question?

- ☐ Not at all helpful
- ☐ Slightly helpful
- ☐ Moderately helpful
- ☐ Very helpful
- ☐ Extremely helpful

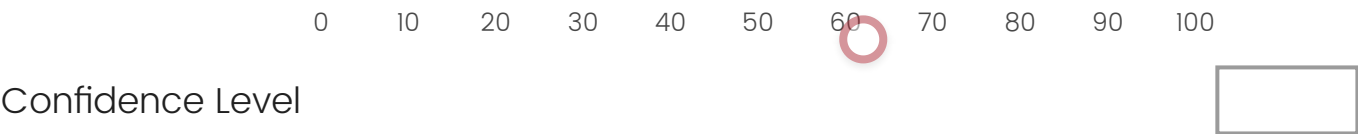
Do you think Table A and Table B are unionable?

Table A		
Title	Artist(s)	Genre
Bohemian Rhapsody	Queen	Rock
Stairway to Heaven	Led Zeppelin	Rock
Billie Jean	Michael Jackson	Pop
Hotel California	Eagles	Rock
Imagine	John Lennon	Soft Rock

Table B		
Alternative Title	Additional Artist	Genre Diversify
Sweet Child O' Mine	Guns N' Roses	Hard Rock
Smells Like Teen Spirit	Nirvana	Grunge
Hey Jude	The Beatles	Rock
Comfortably Numb	Pink Floyd	Progressive Rock
Black Dog	Led Zeppelin	Hard Rock

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your answer to the question?



Do you think Table A and Table B are unionable?

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Hey Jude	The Beatles	Rock
Comfortably Numb	Pink Floyd	Progressive Rock
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LLM Recommendation: Unionable

LLM Reason: Both tables represent songs with corresponding attributes for title, artist, and genre, indicating strong semantic alignment. The column names differ slightly (e.g., "Title" vs. "Alternative Title"), but they refer to equivalent concepts and can be normalized. Structurally, both tables have the same number of columns and compatible data types, making union feasible.

After reviewing the AI suggestion, what is your final answer?

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your final answer?

0 10 20 30 40 50 60 70 80 90 100

Confidence Level

Please briefly explain your final decision and describe whether the AI suggestion changed, confirmed, or did not

affect your reasoning.

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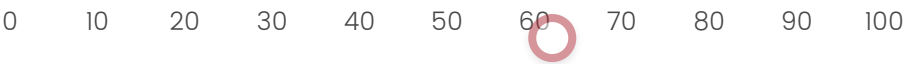
Do you think Table A and Table B are unionable?

Table A		
Party	Year	Leader
Democrat	2020	Tara Lopez
Republican	2018	Joe Jones
Independent	2020	Sarah Peters
Democrat	2018	John Smith
Republican	2020	Mike Hamilton

Table B			
Ideology	Leader	Party	Popularity
Conservative	Mike Hamilton	Republican	High
Liberal	Tara Lopez	Democrat	Low
Progressive	Sarah Peters	Independent	Moderate
Liberal	John Smith	Democrat	High
Conservative	Joe Jones	Republican	Moderate

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your answer to the question?



Confidence Level

Do you think Table A and Table B are unionable?

Table A			Table B			
Party	Year	Leader	Ideology	Leader	Party	Popularity
Democrat	2020	Tara Lopez	Conservative	Mike Hamilton	Republican	High
Republican	2018	Joe Jones	Liberal	Tara Lopez	Democrat	Low
Independent	2020	Sarah Peters	Progressive	Sarah Peters	Independent	Moderate
Democrat	2018	John Smith	Liberal	John Smith	Democrat	High
Republican	2020	Mike Hamilton	Conservative	Joe Jones	Republican	Moderate

LLM Recommendation: Not unionable

LLM Reason: Table A contains three attributes (Party, Year, Leader), while Table B includes four attributes (Ideology, Leader, Party, Popularity), resulting in a structural mismatch. Although there is partial semantic overlap (e.g., Party and Leader), Table B introduces additional dimensions not present in Table A. This difference in schema and attribute meaning prevents a clean, consistent union without significant transformation.

LLM Confidence: 92%

After reviewing the AI suggestion, what is your final answer?

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your final answer?

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Confidence Level

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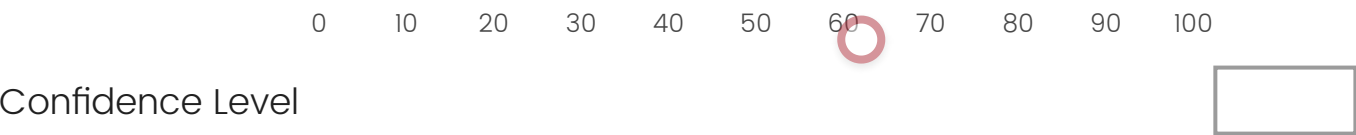
Do you think Table A and Table B are unionable?

Table A		
Calculus	Differentiation, Integration	Focuses on change
Algebra	Equations, identities	Symbols and rules
Geometry	Points, lines, planes, angles	Shapes and sizes
Probability	Events, sample space	Study of chance
Number Theory	Prime numbers, congruences	Properties of numbers

Table B		
Statistics	Mean, median, mode	Data Analysis
Linear Algebra	Vectors, matrices	Algebraic Structures
Combinatorics	Counting, arrangements	Discrete Mathematics
Topology	Connectedness, continuity	Spatial Properties
Logic	Propositions, inference	Foundations of Mathematics

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your answer to the question?



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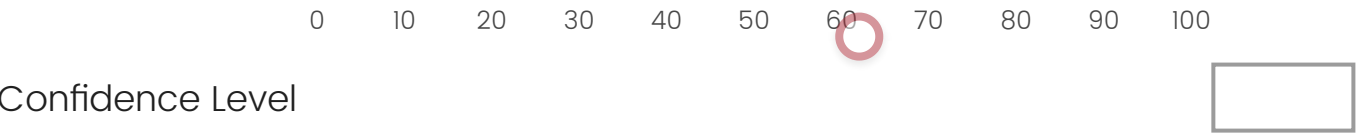
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LLM Recommendation: Unionable

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- ☐ Yes
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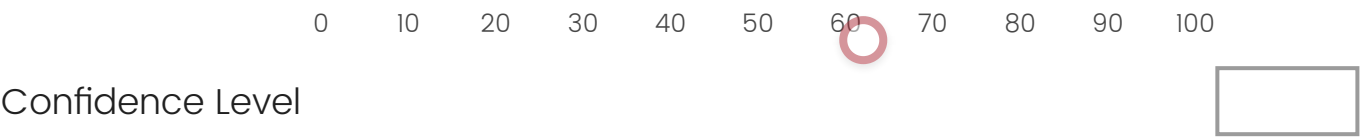
Do you think Table A and Table B are unionable?

Table A		
Concept Name	Control	Scientific Discipline
Gravity	A stationary object	Physics
Acid-Base Reactions	Deionized water	Chemistry
Photosynthesis	A fully lit room	Biology
Plate Tectonics	A location where two tectonic plates meet	Earth Science
Drug Efficacy	A group of patients with the same condition	Medicine

Table B		
Concept Name	Control Conditions	Quantum Physics
Quantum Interference	Coherent light sources	Wave-Particle Duality
Heisenberg Uncertainty Principle	Simultaneous measurements	Measurement Theory
Bell's Theorem	Local hidden variables	Quantum Entanglement
Quantum Computing	Error correction methods	Qubits and Superposition
Schrödinger's Equation	Initial wave function	Quantum Dynamics

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your answer to the question?



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Bell's Theorem	Local hidden variables	Quantum Entanglement
Quantum Computing	Error correction methods	Qubits and Superposition
Schrödinger's Equation	Initial wave function	Quantum Dynamics

LLM Recommendation: Not unionable

LLM Reason: While both tables share a similar structure (concept name and control-related attributes), their domains differ significantly: Table A spans multiple scientific disciplines, whereas Table B is focused specifically on quantum physics. The third column represents different semantic meanings (general discipline vs. specific quantum physics concepts), leading to misalignment. This mismatch in context and

attribute meaning prevents a coherent union.

LLM Confidence: 91%

After reviewing the AI suggestion, what is your final answer?

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your final answer?

0 10 20 30 40 50 60 70 80 90 100

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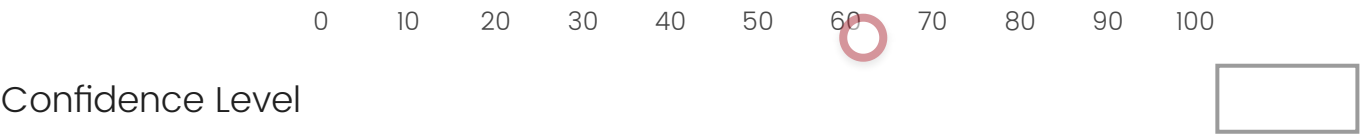
Do you think Table A and Table B are unionable?

Table A		
Ages	Gods	Actions
20th Century	Hera	Ethos
Neoplatonic	Artemis	Form
Classical	Hades	Eros
Ancient	Aphrodite	Nous
21st Century	Poseidon	Kosmos

Table B		
Philosophers	Religion	Process
Locke	Judaism	Dualism
Descartes	Islam	Idealism
Hegel	Buddhism	Realism
Hume	Taoism	Nominalism
Kant	Christianity	Monism

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your answer to the question?



Do you think Table A and Table B are unionable?

Table A			Table B		
Ages	Gods	Actions	Philosophers	Religion	Process
20th Century	Hera	Ethos	Locke	Judaism	Dualism
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Classical	Hades	Eros	Hegel	Buddhism	Realism
Ancient	Aphrodite	Nous	Hume	Taoism	Nominalism
21st Century	Poseidon	Kosmos	Kant	Christianity	Monism

LLM Recommendation: Not unionable

LLM Reason: Although both tables have three columns, they represent entirely different domains: Table A is about mythological or philosophical constructs across ages, while Table B focuses on philosophers, religions, and philosophical processes. There is no semantic alignment between corresponding columns (e.g., Gods vs. Philosophers, Actions vs. Process). This mismatch makes a meaningful union infeasible without redefining the schema.

After reviewing the AI suggestion, what is your final answer?

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your

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Do you think Table A and Table B are unionable?

Table A		
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Plants	Wave-Particle Duality	Irrigation
Energy	Solar Tubes	Dikes

Table B		
Land	Zero Net Energy	Floodplain
Energy	Tubular Skylights	Drainage Basins
Animals	Light Tubes	Channelization
Oceans	Passive Cooling	Reefs
Plants	Daylighting System	Flood Planning

- ☐ Yes
- ☐ No

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Confidence Level

Do you think Table A and Table B are unionable?

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LLM Recommendation: Unionable

LLM Reason: Both tables share the same implicit structure with three columns representing environment category, air-related aspect, and water-related aspect. The entries follow the same schema and data types, even though specific values differ. This strong structural and semantic alignment makes them directly unionable.

After reviewing the AI suggestion, what is your final answer?

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your final answer?

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Confidence Level

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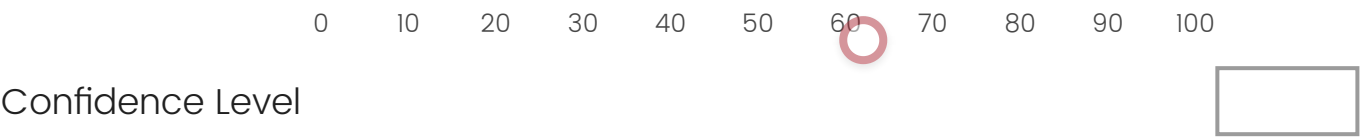
Do you think Table A and Table B are unionable?

Table A		
Processors	Programming	Bluetooth
Intel Core i7	Debugging	Wi-Fi
A9 Chip	Coding	Infrared
HD Output	Conferencing	Wi-Fi
RAID	Backup	Bluetooth

Table B		
Windows	Word	6-Cell
macOS	App	8-Cell
Android	Photoshop	2-Cell
Web-Based	Facetime	Rechargeable
Linux	Terminal	Lithium

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your answer to the question?



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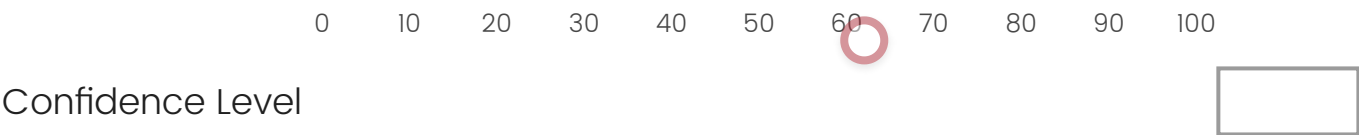
Do you think Table A and Table B are unionable?

Table A		
Assonance	Couplet	Interlinking
Alliteration	Limerick	Blank Verse
Personification	Sestina	Stressed
Enjambment	Quatrain	Metrical Pattern
Metaphor	Ballad	Free Verse

Table B			
Free Verse	Characterization	Dialogue	Iambic Pentameter
Blank Verse	Imagery	Simile	Dramatic Monologue
Lyric	Description	Symbolic Language	Internal Monologue
Epic	Alliteration	Rhyme Scheme	Reflective Monologue
Anapestic	Metaphor	Refrain	Poetic Monologue

- ☐ Yes
- ☐ No

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Epic	Alliteration	Rhyme Scheme	Reflective Monologue
Anapestic	Metaphor	Refrain	Poetic Monologue

LLM Recommendation: Not unionable

LLM Reason: Table A has three columns related to poetic

devices and structures, while Table B has four columns covering broader literary elements and forms, creating a structural mismatch. Additionally, there is no consistent one-to-one semantic alignment between columns. This prevents a meaningful and coherent union.

LLM Confidence: 96%

After reviewing the AI suggestion, what is your final answer?

- ☐ Yes
- ☐ No

On a scale from 0 to 100, how confident are you in your final answer?

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Confidence Level

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